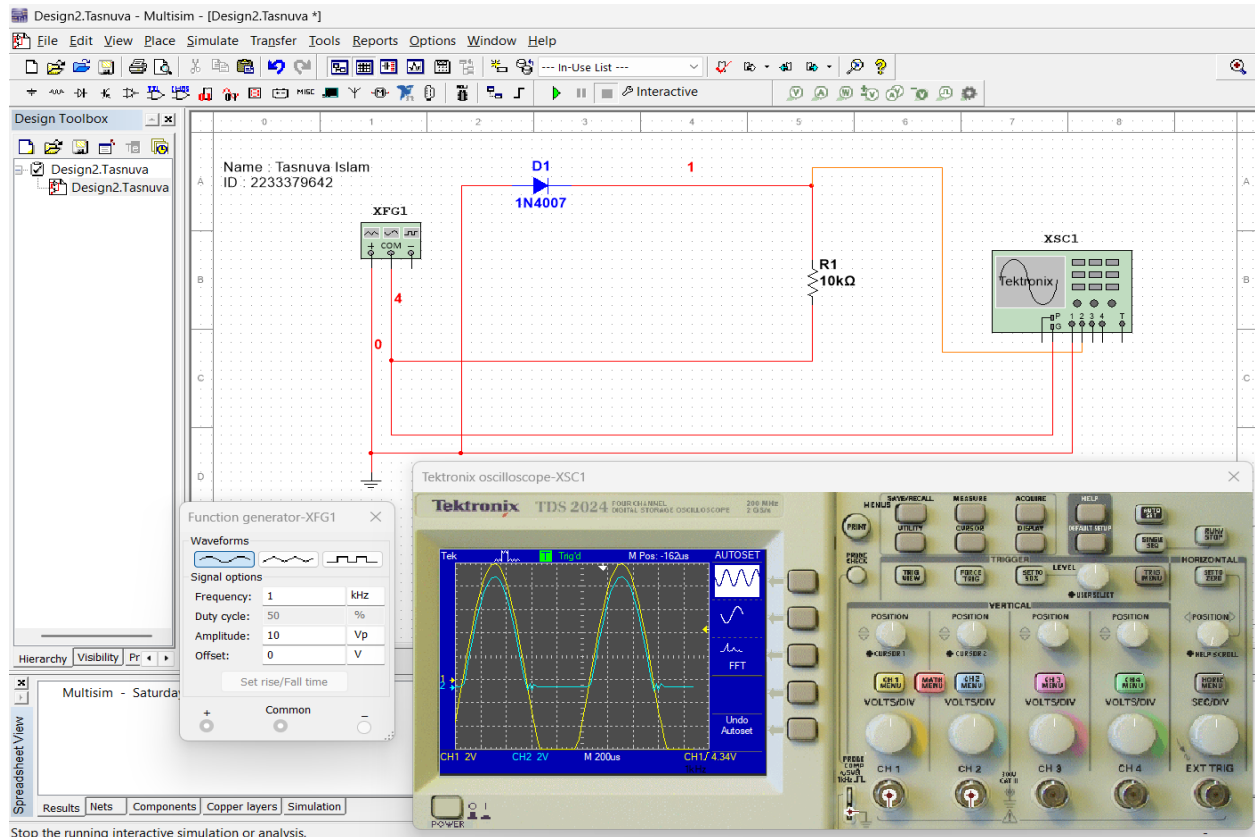


Half- Wave Rectifier Circuit :

Without capacitor :



The screenshot displays the Multisim software interface for a circuit simulation. The main workspace shows a circuit diagram with the following components and connections:

- Function Generator (XFG1):** Configured with a sine wave, 1 kHz frequency, 50% duty cycle, 10 Vp amplitude, and 0 V offset.
- Diode (D1, 1N4007):** Connected in series with the function generator.
- Resistor (R1, 10kΩ):** Connected in series with the diode.
- Capacitor (C1, 0.22μF):** Connected in parallel with the resistor.
- Oscilloscope (XSC1):** Connected to the output of the circuit, displaying two waveforms: a sine wave (CH1) and a rectified sine wave (CH2).

The oscilloscope screen shows the following settings and measurements:

- Model:** Tektronix TDS 2024
- Mode:** AUTOSET
- CH1:** 4.34V, 200us
- CH2:** 2V, 200us
- M Pos:** -16.0us
- Waveform:** Sine wave (CH1) and Rectified Sine wave (CH2)

The Design Toolbox on the left contains the components used in the circuit. The bottom status bar indicates the simulation is running.

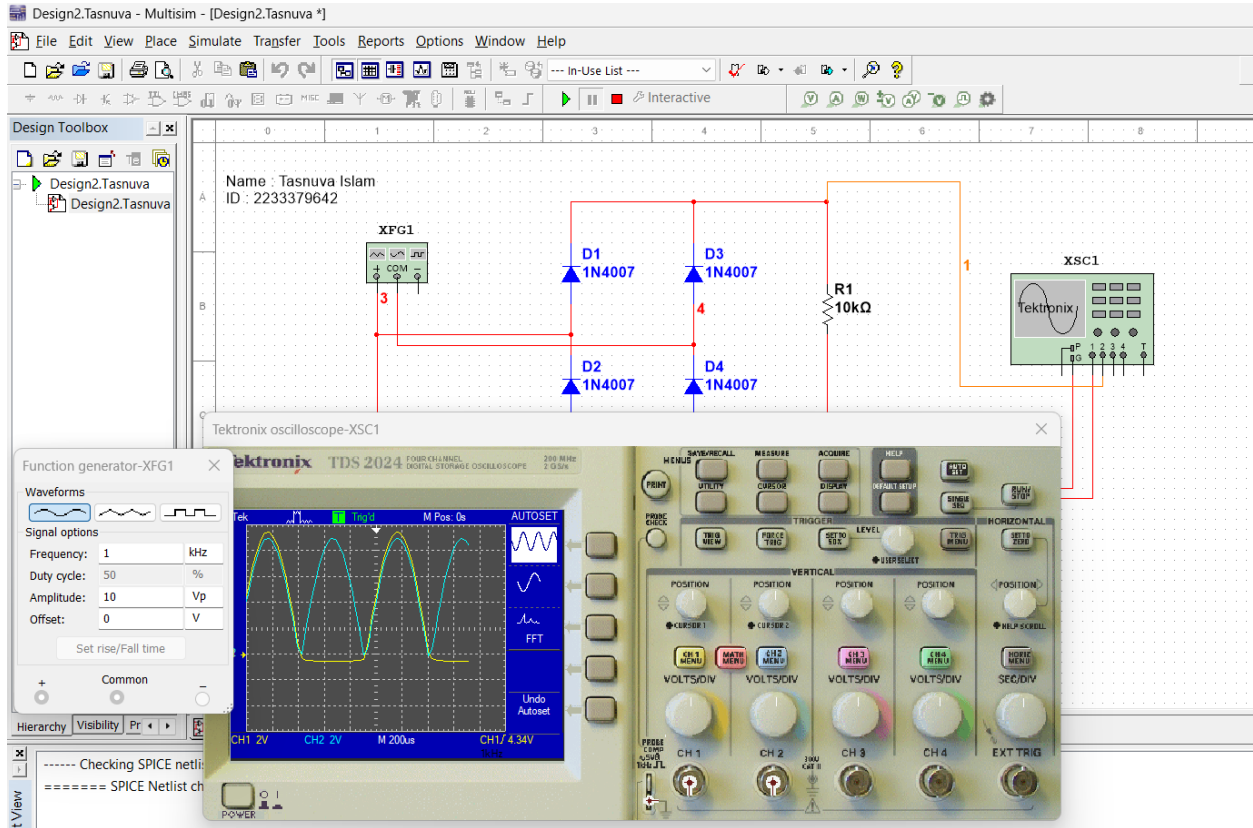
The screenshot displays the Multisim software interface for a circuit simulation. The main workspace shows a circuit diagram with the following components and connections:

- Function Generator (XFG1):** Configured with a sine wave, 1 kHz frequency, 50% duty cycle, 10 Vp amplitude, and 0 V offset. It is connected to the anode of the diode (D1).
- Diode (D1):** A 1N4007 diode connected in series with the function generator.
- Resistor (R1):** A 10 kΩ resistor connected in series with the diode.
- Capacitor (C1):** A 10 μF capacitor connected in parallel with the resistor.
- Oscilloscope (XSC1):** A Tektronix TDS 2024 oscilloscope connected to the output of the circuit (across the capacitor).

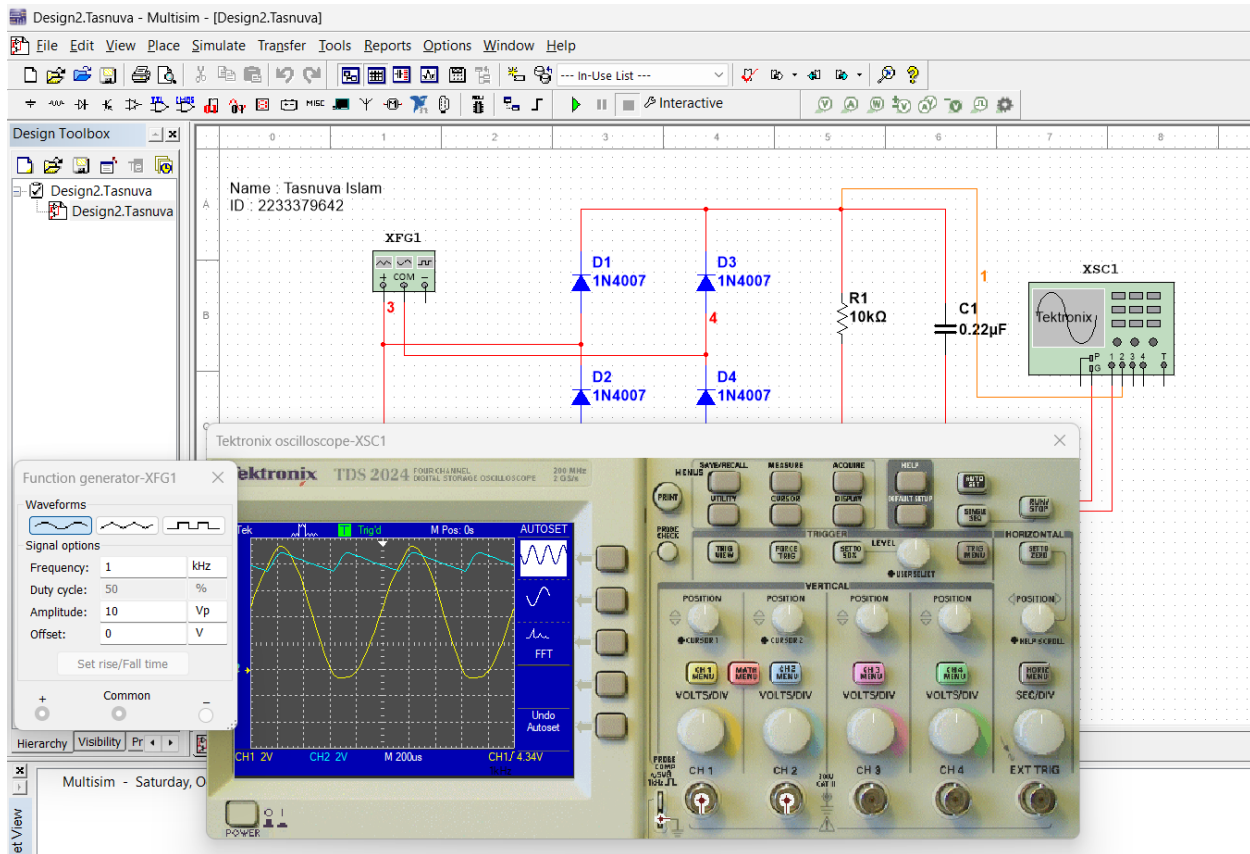
The oscilloscope screen shows a waveform with a peak-to-peak voltage of approximately 10 V and a period of 200 μs. The function generator's settings are displayed in the bottom-left panel, and the oscilloscope's controls are visible in the bottom-right panel.

Full - Wave Rectifier Circuit :

Without capacitor :



With 0.22uF capacitor :



With 10uF capacitor :

